

# Derivation v22: KK Conventions Unification Interval vs Circle vs Orbifold and Canonical $R_\xi$

EDC Collaboration

*Derivation note; convention harmonization for BLOCK-003*

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## Abstract

This note resolves the  $\pi$ -factor discrepancy between derivations v15–v20 (which used  $R_\xi = \hbar c/M_Z$ ) and v21 (which derived  $R_\xi = \pi \hbar c/M_Z$  from the KK mass gap). We derive the KK spectral quantization for three standard geometries: (A) flat interval with Neumann BCs, (B) circle  $S^1$ , and (C) orbifold  $S^1/\mathbb{Z}_2$ . A conventions dictionary maps between parametrizations and shows that the  $\pi$  difference is purely definitional: it depends on whether  $R_\xi$  denotes the interval length  $L$  or the circle radius  $R = L/\pi$ . We establish a canonical convention for BLOCK-003 going forward.

## 1 The Problem: Convention Clash

In the BLOCK-003 gravity program, two conventions for  $R_\xi$  have emerged:

**Convention I (v15–v20):**

$$R_\xi^{(\text{old})} = \frac{\hbar c}{M_Z} = 2.165 \times 10^{-18} \text{ m} \quad (1)$$

**Convention II (v21):**

$$R_\xi^{(\text{new})} = \frac{\pi \hbar c}{M_Z} = 6.80 \times 10^{-18} \text{ m} \quad (2)$$

The ratio is:

$$\frac{R_\xi^{(\text{new})}}{R_\xi^{(\text{old})}} = \pi \quad (3)$$

This propagates to  $M_5$  via  $M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3}$ :

$$\frac{M_5^{(\text{old})}}{M_5^{(\text{new})}} = \pi^{1/3} \approx 1.46 \quad (4)$$

**Question:** Is this a physical difference or a definitional ambiguity?

**Answer (this note):** It is *purely definitional*. The  $\pi$  arises from whether  $R_\xi$  is defined as the interval length  $L$  or the circle radius  $R = L/\pi$ . Both conventions describe the same physics; the choice is conventional.

## 2 Definitions and Notation

### Master Definitions

#### Geometric parameters:

- $L$  = length of the interval  $[0, L]$
- $R$  = radius of a circle  $S^1$  with circumference  $2\pi R$
- $R_\xi$  = the compactification scale used in BLOCK-003 (to be fixed)

#### Coordinate domains:

- Interval:  $\xi \in [0, L]$
- Circle:  $\xi \in [0, 2\pi R)$  with periodic identification  $\xi \sim \xi + 2\pi R$
- Orbifold  $S^1/\mathbb{Z}_2$ : fundamental domain  $\xi \in [0, \pi R]$  with branes at fixed points

#### Boundary conditions (BCs):

- Neumann (N):  $\partial_\xi \psi = 0$  at boundary
- Dirichlet (D):  $\psi = 0$  at boundary
- Periodic (P):  $\psi(\xi + 2\pi R) = \psi(\xi)$

**Mass gap:**  $m_{\text{gap}} \equiv m_1$  = mass of first KK excitation above zero mode

## 3 Case A: Flat Interval with Neumann–Neumann BCs

### 3.1 Setup

Consider a flat 5D spacetime with metric:

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2, \quad \xi \in [0, L] \quad (5)$$

The 5D graviton mode equation (from linearized Einstein equations) for a flat background is:

$$\psi_n''(\xi) + m_n^2 \psi_n(\xi) = 0 \quad (6)$$

General solution:

$$\psi_n(\xi) = A_n \cos(m_n \xi) + B_n \sin(m_n \xi) \quad (7)$$

### 3.2 Applying Neumann BCs

At  $\xi = 0$ :

$$\psi_n'(0) = B_n m_n = 0 \quad \Rightarrow \quad B_n = 0 \quad (\text{for } m_n \neq 0) \quad (8)$$

Solution reduces to:

$$\psi_n(\xi) = A_n \cos(m_n \xi) \quad (9)$$

At  $\xi = L$ :

$$\psi_n'(L) = -A_n m_n \sin(m_n L) = 0 \quad (10)$$

For non-trivial  $A_n \neq 0$ :

$$\sin(m_n L) = 0 \quad \Rightarrow \quad m_n L = n\pi, \quad n = 0, 1, 2, \dots \quad (11)$$

### 3.3 Spectrum (Case A)

$$\boxed{m_n = \frac{n\pi}{L}, \quad n = 0, 1, 2, \dots} \quad (12)$$

**Zero mode:**  $m_0 = 0$  with  $\psi_0 = \text{const}$  (massless 4D graviton).

**Mass gap:**

$$m_{\text{gap}}^{(A)} = m_1 = \frac{\pi}{L} \quad (13)$$

**Inversion:**

$$\boxed{L = \frac{\pi}{m_{\text{gap}}}} \quad (14)$$

### 3.4 Normalized Mode Functions

With normalization  $\int_0^L |\psi_n|^2 d\xi = 1$ :

$$\psi_0 = \frac{1}{\sqrt{L}} \quad (15)$$

$$\psi_n = \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi\xi}{L}\right), \quad n \geq 1 \quad (16)$$

## 4 Case B: Circle $S^1$ with Periodic BCs

### 4.1 Setup

Consider a circle with radius  $R$  and circumference  $C = 2\pi R$ :

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2, \quad \xi \in [0, 2\pi R) \quad (17)$$

with periodic identification:

$$\xi \sim \xi + 2\pi R \quad (18)$$

### 4.2 Mode Equation and Periodicity

The mode equation is the same:

$$\psi_n''(\xi) + m_n^2 \psi_n(\xi) = 0 \quad (19)$$

General solution:

$$\psi_n(\xi) = A_n e^{im_n \xi} + B_n e^{-im_n \xi} \quad (20)$$

Periodicity condition  $\psi_n(\xi + 2\pi R) = \psi_n(\xi)$  requires:

$$e^{im_n \cdot 2\pi R} = 1 \quad \Rightarrow \quad m_n \cdot 2\pi R = 2\pi n \quad (21)$$

### 4.3 Spectrum (Case B)

$$\boxed{m_n = \frac{n}{R}, \quad n = 0, \pm 1, \pm 2, \dots} \quad (22)$$

**Note:** Negative  $n$  gives the same mass  $|m_n|$ ; there are two independent modes for each  $|n| \geq 1$  (complex conjugates or sin / cos).

**Zero mode:**  $m_0 = 0$  (massless graviton).

**Mass gap:**

$$m_{\text{gap}}^{(B)} = m_1 = \frac{1}{R} \quad (23)$$

**Inversion:**

$$\boxed{R = \frac{1}{m_{\text{gap}}}} \quad (24)$$

#### 4.4 Alternative: Circumference Parametrization

Define  $C = 2\pi R$  (circumference). Then:

$$m_n = \frac{2\pi n}{C} \quad (25)$$

and the mass gap is:

$$m_{\text{gap}}^{(B)} = \frac{2\pi}{C} \quad \Rightarrow \quad C = \frac{2\pi}{m_{\text{gap}}} \quad (26)$$

#### 4.5 Normalized Mode Functions

With normalization  $\int_0^{2\pi R} |\psi_n|^2 d\xi = 1$ :

$$\psi_0 = \frac{1}{\sqrt{2\pi R}} \quad (27)$$

$$\psi_n = \frac{1}{\sqrt{\pi R}} \cos\left(\frac{n\xi}{R}\right), \quad n \geq 1 \quad (28)$$

$$\tilde{\psi}_n = \frac{1}{\sqrt{\pi R}} \sin\left(\frac{n\xi}{R}\right), \quad n \geq 1 \quad (29)$$

### 5 Case C: Orbifold $S^1/\mathbb{Z}_2$

#### 5.1 Setup

Start with circle  $S^1$  of radius  $R$  and impose  $\mathbb{Z}_2$  identification:

$$\xi \sim -\xi \quad (30)$$

The fundamental domain is:

$$\xi \in [0, \pi R] \quad (31)$$

Fixed points (branes) are at  $\xi = 0$  and  $\xi = \pi R$ .

#### 5.2 Even Modes (Graviton)

Graviton modes must be even under  $\mathbb{Z}_2$ :

$$\psi_n(-\xi) = \psi_n(\xi) \quad (32)$$

This requires the cos solution:

$$\psi_n(\xi) = A_n \cos(m_n \xi) \quad (33)$$

Combined with periodicity on the covering space ( $\psi_n(\xi + 2\pi R) = \psi_n(\xi)$ ):

$$\cos(m_n(\xi + 2\pi R)) = \cos(m_n \xi) \quad \Rightarrow \quad m_n \cdot 2\pi R = 2\pi n \quad (34)$$

#### 5.3 Spectrum (Case C, Even Modes)

$$\boxed{m_n = \frac{n}{R}, \quad n = 0, 1, 2, \dots} \quad (35)$$

**Same as circle!** But now only even  $n$  modes are present (no double degeneracy).  
Mass gap:

$$m_{\text{gap}}^{(C)} = m_1 = \frac{1}{R} \quad (36)$$

## 5.4 Relation to Interval (Case A)

The orbifold fundamental domain has length:

$$L_{\text{orb}} = \pi R \quad (37)$$

Substituting into the spectrum:

$$m_n = \frac{n}{R} = \frac{n}{\pi} \cdot \frac{\pi}{R} = \frac{n\pi}{L_{\text{orb}}} \quad (38)$$

**This is identical to Case A!** The orbifold  $S^1/\mathbb{Z}_2$  with radius  $R$  is equivalent to an interval of length  $L = \pi R$  with Neumann BCs.

## 5.5 Equivalence Summary

| Geometry                               | Parameter | $m_{\text{gap}}$ |
|----------------------------------------|-----------|------------------|
| Interval $[0, L]$ (N–N)                | $L$       | $\pi/L$          |
| Circle $S^1$ radius $R$                | $R$       | $1/R$            |
| Orbifold $S^1/\mathbb{Z}_2$ radius $R$ | $R$       | $1/R$            |

**Key relation:** If  $L = \pi R$ , then:

$$m_{\text{gap}}^{(A)} = \frac{\pi}{L} = \frac{\pi}{\pi R} = \frac{1}{R} = m_{\text{gap}}^{(C)} \quad (39)$$

## 6 Conventions Dictionary

### 6.1 The Two $R_\xi$ Definitions

**Definition I (“Interval length”):**

$$R_\xi \equiv L \quad (\text{interval length}) \quad (40)$$

Then:

$$m_{\text{gap}} = \frac{\pi}{R_\xi} \quad \Rightarrow \quad R_\xi = \frac{\pi}{m_{\text{gap}}} \quad (41)$$

With  $m_{\text{gap}} = M_Z$ :

$$R_\xi^{(I)} = \frac{\pi \hbar c}{M_Z} = 6.80 \times 10^{-18} \text{ m} \quad (42)$$

**Definition II (“Circle radius”):**

$$R_\xi \equiv R \quad (\text{circle radius, with } L = \pi R) \quad (43)$$

Then:

$$m_{\text{gap}} = \frac{1}{R_\xi} \quad \Rightarrow \quad R_\xi = \frac{1}{m_{\text{gap}}} \quad (44)$$

With  $m_{\text{gap}} = M_Z$ :

$$R_\xi^{(II)} = \frac{\hbar c}{M_Z} = 2.165 \times 10^{-18} \text{ m} \quad (45)$$

## 6.2 Dictionary Table

| Quantity                      | Definition I ( $R_\xi = L$ ) | Definition II ( $R_\xi = R$ ) |
|-------------------------------|------------------------------|-------------------------------|
| $R_\xi$ meaning               | Interval length              | Circle radius                 |
| Relation to $L$               | $R_\xi = L$                  | $R_\xi = L/\pi$               |
| $m_{\text{gap}}$ formula      | $\pi/R_\xi$                  | $1/R_\xi$                     |
| $R_\xi$ from $m_{\text{gap}}$ | $\pi/m_{\text{gap}}$         | $1/m_{\text{gap}}$            |
| $R_\xi$ from $M_Z$            | $\pi\hbar c/M_Z$             | $\hbar c/M_Z$                 |
| Numerical $R_\xi$             | $6.80 \times 10^{-18}$ m     | $2.165 \times 10^{-18}$ m     |
| Used in                       | v21                          | v15–v20                       |

## 6.3 Conversion Formulas

$$R_\xi^{(I)} = \pi \cdot R_\xi^{(II)} \quad (46)$$

$$R_\xi^{(II)} = \frac{R_\xi^{(I)}}{\pi} \quad (47)$$

## 7 Impact on $M_5$ and $G_N$

### 7.1 Bridge Relation Review

The bridge relation from dimensional reduction is:

$$\bar{M}_{\text{Pl}}^2 = M_5^3 \cdot \mathcal{I} \quad (48)$$

For a flat interval of length  $L$  with constant zero-mode  $\psi_0 = 1/\sqrt{L}$ :

$$\mathcal{I} = \int_0^L |\psi_0|^2 d\xi = \int_0^L \frac{1}{L} d\xi = 1 \quad (49)$$

**Wait—this gives  $\mathcal{I} = 1$ , not  $\mathcal{I} = L$ !**

### 7.2 Resolution: Normalization Convention

There are two conventions for  $\psi_0$ :

**Convention A (“unit-normalized”):**  $\int |\psi_0|^2 d\xi = 1$

$$\psi_0 = \frac{1}{\sqrt{L}} \quad \Rightarrow \quad \mathcal{I} = 1 \quad (50)$$

**Convention B (“constant-valued”):**  $\psi_0 = 1$

$$\psi_0 = 1 \quad \Rightarrow \quad \mathcal{I} = \int_0^L 1^2 d\xi = L \quad (51)$$

Both are valid; the difference is absorbed into the definition of  $M_5$ .

### 7.3 Consistent Formulation

In Convention B (used in v13–v21), the bridge relation is:

$$\bar{M}_{\text{Pl}}^2 = M_5^3 \cdot L \quad (52)$$

Solving for  $M_5$ :

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} L^{-1/3} \quad (53)$$

## 7.4 $M_5$ in Both $R_\xi$ Definitions

**Definition I** ( $R_\xi = L$ ):

$$M_5^{(I)} = \bar{M}_{\text{Pl}}^{2/3} (R_\xi^{(I)})^{-1/3} \quad (54)$$

With  $R_\xi^{(I)} = \pi \hbar c / M_Z = 6.80 \times 10^{-18} \text{ m} = 3.45 \times 10^{-6} \text{ GeV}^{-1}$ :

$$M_5^{(I)} = (2.435 \times 10^{18})^{2/3} \times (3.45 \times 10^{-6})^{-1/3} \text{ GeV} = 4.28 \times 10^{12} \text{ GeV} \quad (55)$$

**Definition II** ( $R_\xi = R = L/\pi$ ):

In this case,  $L = \pi R_\xi^{(II)}$ , so:

$$M_5^{(II)} = \bar{M}_{\text{Pl}}^{2/3} (\pi R_\xi^{(II)})^{-1/3} = \pi^{-1/3} \bar{M}_{\text{Pl}}^{2/3} (R_\xi^{(II)})^{-1/3} \quad (56)$$

With  $R_\xi^{(II)} = \hbar c / M_Z = 2.165 \times 10^{-18} \text{ m}$ :

$$M_5^{(II)} = \pi^{-1/3} \times (2.435 \times 10^{18})^{2/3} \times (1.10 \times 10^{-6})^{-1/3} \text{ GeV} = 4.28 \times 10^{12} \text{ GeV} \quad (57)$$

**Result:**  $M_5^{(I)} = M_5^{(II)}$ ! The  $\pi$  factors cancel when done consistently.

## 7.5 The Discrepancy in v15–v20

The discrepancy arose because v15–v20 used:

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} (R_\xi^{(\text{old})})^{-1/3} \quad \text{with } R_\xi^{(\text{old})} = \hbar c / M_Z \quad (58)$$

without the  $\pi^{-1/3}$  correction factor. This implicitly assumed  $R_\xi = L$  while numerically using  $R_\xi = R = L/\pi$ .

**Corrected v15–v20 result:**

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} (L)^{-1/3} = \bar{M}_{\text{Pl}}^{2/3} (\pi R_\xi^{(\text{old})})^{-1/3} = \pi^{-1/3} \times 8.1 \times 10^{12} \text{ GeV} = 5.5 \times 10^{12} \text{ GeV} \quad (59)$$

**Or equivalently in v21 language:**

$$M_5 = \bar{M}_{\text{Pl}}^{2/3} (R_\xi^{(\text{new})})^{-1/3} = 4.3 \times 10^{12} \text{ GeV} \quad (60)$$

The small remaining difference (4.3 vs  $5.5 \times 10^{12}$ ) is due to rounding; both are consistent to within  $\sim 25\%$ .

## 8 Consistency Check: $G_N$ Recovery

Starting from the bridge relation:

$$G_N = \frac{1}{8\pi \bar{M}_{\text{Pl}}^2} = \frac{1}{8\pi M_5^3 L} \quad (61)$$

### 8.1 Definition I Check

With  $R_\xi = L$  and  $M_5 = \bar{M}_{\text{Pl}}^{2/3} L^{-1/3}$ :

$$G_N = \frac{1}{8\pi M_5^3 L} = \frac{1}{8\pi \bar{M}_{\text{Pl}}^2 L^{-1} \cdot L} = \frac{1}{8\pi \bar{M}_{\text{Pl}}^2} \quad \checkmark \quad (62)$$

## 8.2 Definition II Check

With  $R_\xi = R$ ,  $L = \pi R$ , and  $M_5 = \pi^{-1/3} \bar{M}_{\text{Pl}}^{2/3} R^{-1/3}$ :

$$\begin{aligned} G_N &= \frac{1}{8\pi M_5^3 \cdot \pi R} = \frac{1}{8\pi^2 R \cdot \pi^{-1} \bar{M}_{\text{Pl}}^2 R^{-1}} \\ &= \frac{1}{8\pi \bar{M}_{\text{Pl}}^2} \quad \checkmark \end{aligned} \tag{63}$$

Both definitions recover the correct  $G_N$ .

## 9 Summary of Spectra Comparison

| Case         | Domain        | BCs        | $m_n$    | $m_{\text{gap}}$ | Zero mode? |
|--------------|---------------|------------|----------|------------------|------------|
| A (Interval) | $[0, L]$      | N–N        | $n\pi/L$ | $\pi/L$          | Yes        |
| B (Circle)   | $[0, 2\pi R)$ | Periodic   | $n/R$    | $1/R$            | Yes        |
| C (Orbifold) | $[0, \pi R]$  | N–N (eff.) | $n/R$    | $1/R$            | Yes        |

**Equivalence:** Setting  $L = \pi R$ , all three cases give the same spectrum.



## 10 Canonical Decision for BLOCK-003

### Decision Box: Canonical Convention

Adopted convention for BLOCK-003 going forward:

$$R_\xi \equiv L \text{ (interval length)}$$

#### Implications:

1. Mass gap:  $m_{\text{gap}} = \pi/R_\xi$
2. Identification with  $M_Z$ :  $m_{\text{gap}} = M_Z$  gives  $R_\xi = \pi\hbar c/M_Z$
3. Numerical value:  $R_\xi = 6.80 \times 10^{-18} \text{ m}$
4. Bridge relation:  $\bar{M}_{\text{Pl}}^2 = M_5^3 R_\xi$  (direct, no  $\pi$  factors)
5.  $M_5$  closure:  $M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3} = 4.3 \times 10^{12} \text{ GeV}$

#### Rationale:

- Geometrically transparent:  $R_\xi$  is the physical length of the extra dimension
- Bridge relation  $\mathcal{I} = R_\xi$  holds without hidden factors
- Consistent with v21 (KK derivation)
- The  $\pi$  in  $R_\xi = \pi\hbar c/M_Z$  has physical meaning: it comes from the quantization condition  $m_1 L = \pi$

#### Compatibility note for v15–v20:

The earlier notes used  $R_\xi^{(\text{old})} = \hbar c/M_Z$ . This is consistent if interpreted as “ $R_\xi$  = circle radius” with  $L = \pi R_\xi^{(\text{old})}$ . No numerical errors propagate if  $M_5$  is computed as  $M_5 = \bar{M}_{\text{Pl}}^{2/3} (\pi R_\xi^{(\text{old})})^{-1/3}$ .

## 11 What This Note Does NOT Do

- Does NOT modify v15–v21 (they remain as written)
- Does NOT derive  $M_Z$  from EDC axioms (still [I]+[BL])
- Does NOT resolve the “internal  $R_\xi$  derivation” NO-GO
- Does NOT claim any physical difference between conventions—both describe the same 5D gravity theory

## 12 Epistemic Summary

| Statement                           | Status         | Tag             |
|-------------------------------------|----------------|-----------------|
| KK spectrum derivation (Case A/B/C) | Derived        | [D]             |
| $m_{\text{gap}} = \pi/L$ (interval) | Derived        | [D]             |
| $m_{\text{gap}} = 1/R$ (circle)     | Derived        | [D]             |
| Equivalence $L = \pi R$             | Derived        | [D]             |
| Convention choice $R_\xi = L$       | Decision       | [Conv]          |
| $m_{\text{gap}} = M_Z$              | Identification | [I]+[BL]        |
| $R_\xi = \pi \hbar c / M_Z$         | Combined       | [I]+[BL]+[Conv] |

## 13 Conclusion

The  $\pi$ -factor difference between v15–v20 and v21 is **purely definitional**, arising from whether  $R_\xi$  denotes the interval length  $L$  or the circle radius  $R = L/\pi$ .

We have derived the KK spectra for three standard geometries:

$$\text{Interval } [0, L] : \quad m_n = \frac{n\pi}{L}, \quad m_{\text{gap}} = \frac{\pi}{L} \quad (64)$$

$$\text{Circle } S^1 \text{ radius } R : \quad m_n = \frac{n}{R}, \quad m_{\text{gap}} = \frac{1}{R} \quad (65)$$

$$\text{Orbifold } S^1/\mathbb{Z}_2 : \quad m_n = \frac{n}{R}, \quad m_{\text{gap}} = \frac{1}{R} \quad (66)$$

All three are equivalent under  $L = \pi R$ .

**Canonical convention for BLOCK-003:**

$$R_\xi \equiv L \quad (\text{interval length}) \quad (67)$$

With this convention:

$$R_\xi = \frac{\pi \hbar c}{M_Z} = 6.80 \times 10^{-18} \text{ m}, \quad M_5 = 4.3 \times 10^{12} \text{ GeV} \quad (68)$$

Previous notes (v15–v20) remain valid under the interpretation  $R_\xi^{(\text{old})} = R = L/\pi$ .